P8107 FINAL EXAM

Fall 2023

Name:

UNI:

For this exam you allowed up to four sheets of paper (front and back). No other reference materials may be used. You have 2 hour and 50 minutes to complete this exam. Be sure to show all your work. You may attach additional pages to your exam paper if needed.

Honor code:
I have not and will not give or receive aid in this examination nor have I concealed any violation of the University Honor Code.

Sign here: ________________________________

1. (10 points) If $E[X] = 3$, $Var(X) = 4$, $E[Y] = 1$, $Var(Y) = 2$, and $Cov(X, Y) = -1$, find the expected value and the variance of $2Y - 4X + 1$.

2. (15 points) A random variable Y has density $f(y) = \frac{1}{y\sqrt{2\pi}} e^{-(\log y)^2/2}$ for $y > 0$ (and 0 otherwise), find the distribution of $U = \log Y$.

3. (15 points) A study is being planned to estimate the mean birth weight of infants born full term to teen-aged mothers. A previous large-scale study estimated mean birth weight to all mothers to be 3.4 kg with a standard deviation of 0.5 kg. How many women 19 years of age and under should be enrolled in the study if investigators want a 95% confidence interval of the mean birth weight to have width no more than 0.2 kg? What assumptions are you making in your calculations? Comment briefly on whether they seem appropriate in this situation.

4. (20 points) Random variables $Y_1, Y_2, \ldots, Y_n$ are iid $U(\theta, \theta + 1)$. It is of interest to estimate θ.
   a. Is $\bar{Y}$ an unbiased estimator of θ? Is it consistent for θ?
   b. Is $Y_{(1)}$ (the smallest order statistic of $Y_1, Y_2, \ldots, Y_n$) an unbiased estimator of θ? Is it consistent for θ?

5. (20 Points) Random variables $Y_1, Y_2, \ldots, Y_n \sim$ iid $N(0, \sigma^2)$ and interest is in estimating σ^2.
    a. Find a pivotal quantity based on $\sum_{i=1}^{n} Y_i^2$.
    b. Use the pivotal quantity in (a) to construct a 95% confidence interval for σ^2. [NOTE: If you can't solve (a), you can "trade in" some points to get the answer for (a) so that you can answer (b).]

6. (20 points) Random variables $Y_1, Y_2, \dots Y_n$ are iid with density $f(y) = (2\beta + 2)y^{2\beta+1}$ for $0 < y < 1$ and for $-\frac{1}{2} < \beta < \infty$.
    a. Find the maximum likelihood estimator for β.
    b. Using the (large sample) approximation for the variance of an MLE, find an (approximate) 95% confidence interval for β.